

A swinging arm calibration method for low velocity hot-wire probe calibration

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Abstract. This paper describes a swinging arm calibration facility for the calibration of hot-wire probes at low velocities. During the motion of the probe the anemometer signal is recorded digitally using pre-set angle positions for the start and stop of the data acquisition. The evaluation method described in this paper is based on an integral procedure over a fixed, known probe path length. Tests in the velocity range 1 to 6 m/s demonstrated a high degree of repeatability of the calibration results. Furthermore, the test facilities can easily be adjusted for other low velocity ranges.

1 Introduction

All practical hot-wire and hot-film probes are secondary devices, i.e. they require calibration before their use. Both types of probes respond primarily to the velocity of the fluid in which they are placed and this is the parameter dependence which is considered in this paper. At moderate and high velocity the probe calibration is carried out either (a) in situ against a known velocity field evaluated usually by a pitot-static tube or (b) in a special calibration facility, e.g. the DISA nozzle calibration facility, in which the velocity at the probe position may be evaluated from the pressure drop across the nozzle, as shown in Khan et al. (1987). Both methods have been shown to give consistent results provided that accurate measurements can be obtained for the related pressure difference.

However, when the velocity becomes smaller than about 4 m/s then the value of ΔP becomes so small (< 1 mm H_2O) that accurate results cannot be obtained by the above mentioned methods. In such cases it may be tempting to extend calibration relationships obtained at higher velocity to lower velocities, but results obtained by Bruun et al. (1987) have shown that such an approach may lead to unacceptable errors. Thus at low velocity special calibration methods have to be employed as outlined e.g. by Bradshaw (1975) and Perry (1982). One of these methods relies on a known motion of the calibration fluid or of the probe itself. Several investigations of this type utilizing the linear motion of a probe mounted on a sled have been reported in the velocity

range 0 to 3 m/s (Aydin and Leutheusser 1980; Tabatabai et al. 1986; Tsanis 1987). The details of these investigations centre around procedures for obtaining quiescent fluid and the development of control systems to ensure a constant velocity of the sled, during the calibration time period ΔT for each calibration point.

The independent results of Aydin and Leutheusser (1980) and Tsanis (1987) has shown that the direct (E , V) calibration data can, with a very high degree of accuracy, be approximated by a simple power law

$$E^2 = A + BV^n \quad (1)$$

provided that the investigated velocity range $V_{\max} - V_{\min}$ is restricted to a moderate value of the ratio $V_{\max}/V_{\min}$, typically 10 or less. The above mentioned investigations have shown that with this velocity range restriction a good approximation can be obtained using Eq. (1), with the value of the exponent n depending on the velocity range:

$$\begin{aligned} 0 < V < 0.2 \text{ m/s:} & \quad n = 1.0; \\ 0.05 \text{ m/s} < V < 0.7 \text{ m/s:} & \quad n \cong 0.6-0.8; \\ 0.7 \text{ m/s} < V < 3 \text{ m/s:} & \quad n = 0.4-0.41. \end{aligned}$$

This curvefit principle is utilized in the proposed integral method, and the value of $n \cong 0.43-0.44$ obtained in our velocity range $1 \text{ m/s} < V < 6.5 \text{ m/s}$ agrees closely with these results.

The objective of the proposed new integral calibration method, is to present a calibration method which will produce a comparable accuracy, using a simpler system and reduced calibration time. Such a method can be developed if modern digital techniques are applied to a single sweep using a variable probe velocity. In principle during such a traverse it is possible to record simultaneously both the anemometer output $E(t)$ and the probe position $S(t)$. Knowing $S(t)$ one can then evaluate $V(t)$. By applying a selection procedure to the related simultaneous $E(t)$, $V(t)$ time histories it is possible to select (E , V) values spaced evenly over the required velocity range, from which the required calibration relationship can be evaluated. The disadvantage of this

procedure is that each set of (E, V) values correspond to a single reading, instead of the conventional time average values over a time period ΔT . Consequently, such a method will be sensitive to probe vibrations, and to digital quantization errors caused by the n bit nature of all digital A/D converters. To minimize these errors, the new variable velocity single sweep method is combined with an integral time method.

2 Theory of new calibration method

At low velocities, considerable difficulties are often experienced in obtaining accurate values for the flow velocity V as discussed in the previous section. The proposed method demonstrates that this problem can be overcome by using an integral technique. In this method we shall rewrite Eq. (1) in the form

$$E^2 = KE_0^2 + BV^n \quad (2)$$

where E_0 is the measured voltage at zero flow. Detailed investigations by several investigators, including Khan and Bruun (1986) has shown that if $K \cong 0.8$ then the accuracy of a curvefit using Eq. (2) is for all practical purposes identical to the curvefit obtained for Eq. (1) using a simultaneous solution for A, B and n . In this investigation we shall therefore set $K = 0.8$.

In the current calibration method the fluid velocity was zero. Consequently the velocity seen by the probe was the velocity of the moving probe, and the probe and the probe path were selected so that the probe velocity V was perpendicular to the wire element.

From simple kinematics the relationship between the probe velocity V and the length S on the probe path can be expressed as

$$S = \int_0^t V dt. \quad (3)$$

If one solves Eq. (2) for V and inserts this inverted equation into Eq. (3) then one obtains

$$S = \int_0^t \frac{1}{B^{1/n}} [E^2 - KE_0^2]^{1/n} dt = \frac{1}{B^{1/n}} \int_0^t [E^2 - KE_0^2]^{1/n} dt. \quad (4)$$

To utilize the proposed method it is essential that two tests are carried out for the same distance S . Denoting the two time records as $E_1(t)$ and $E_2(t)$ we have

$$S = \frac{1}{B^{1/n}} \int_0^{t_1} [E_1^2(t) - KE_0^2]^{1/n} dt \quad (5)$$

and

$$S = \frac{1}{B^{1/n}} \int_0^{t_2} [E_2^2(t) - KE_0^2]^{1/n} dt. \quad (6)$$

As the distance S is the same for both tests, and as the term $\frac{1}{B^{1/n}}$ is constant it follows that

$$\int_0^{t_1} [E_1^2(t) - KE_0^2]^{1/n} dt = \int_0^{t_2} [E_2^2(t) - KE_0^2]^{1/n} dt. \quad (7)$$

If we set $E^2(t) - KE_0^2 = E^*(t)$ then Eq. (7) can be expressed as

$$\int_0^{t_1} [E_1^*(t)]^{1/n} dt = \int_0^{t_2} [E_2^*(t)]^{1/n} dt. \quad (8)$$

This equation is observed to contain only one unknown parameter n . To solve Eq. (8) for n , a function Y was defined as

$$Y(n) = \int_0^{t_1} [E_1^*(t)]^{1/n} dt - \int_0^{t_2} [E_2^*(t)]^{1/n} dt. \quad (9)$$

Evaluating Y as a function of n , gave in all cases a curve of the shape shown in Fig. 1 (representing the test A result, Sect. 4). The correct value of n corresponding to $Y=0$ was identified from the following search method: two values of n were chosen so that the real value lies between them (say 0.4 and 0.5) and the corresponding values of Y were calculated (Fig. 1). Drawing a straight line between these two points identified the point m ($n=m, Y=0$). Then the value of Y corresponding to $n=m$ was calculated and the straight line principle applied again to identify point p . This procedure was continued until the value of Y was less than some pre-specified value w . The program was tested with different sets of data, and it was found that for value of $w \leq 0.0001$ there was no observable difference in the evaluated values of n . Then having evaluated n by this method the corresponding value of B was calculated from either Eq. (5) or (6).

3 Test facility and test procedure

In principle the proposed technique can be applied either to a linear or a circular motion. To demonstrate the feasibility

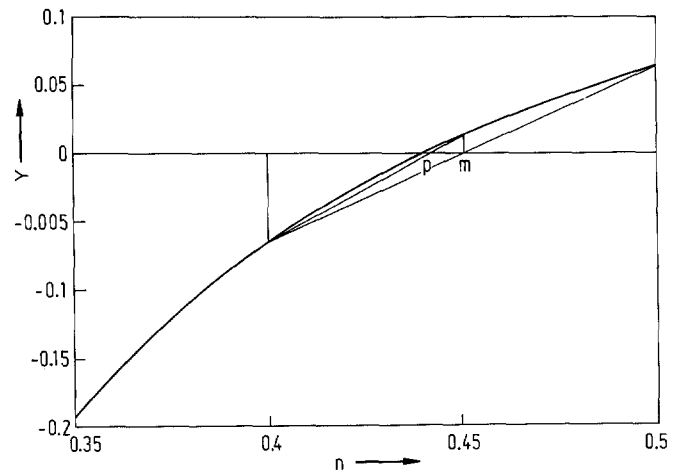


Fig. 1. The $Y(n)$ relationship, Eq. (9), and search procedure (test A data)

of the method a swinging arm test facility was used with the probe at the end of a vertical swinging arm in a gravitational field. The swinging arm test rig is illustrated in Fig. 2. As can be seen the test rig consists of a swinging arm fixed to a rotating shaft mounted on a stand. To minimise the friction in the system and thus reduce vibrations which would contaminate the anemometer signal high quality cylindrical roller bearings were placed at each end of the shaft. Also one half of the shaft was hollow to permit cable connection from the probe to the anemometer set.

As explained in Sect. 2, the method depends on the use of, and the identification of, a constant length S of the probe path. This objective was achieved by placing a disk on one end of the rotating shaft as shown in Fig. 2. This disk had two machined slots near the periphery, placed an angle $\Delta\theta$ apart. This disk (and the slots) run through a compact slotted photo cell unit, containing an infra-red source and integrated photo detector. As slit S1 passes the photo cell it activates the digital sampling of the output from the hot-wire anemometer, which will continue until the second slit S2 passes the photo cell, which stops more samples being acquired. The photo cell was programmed to switch sampling on and off only once because the arm swings like a pendulum causing the slits to rotate past the photo cell again.

As explained in the following, the proposed method requires different mountings of the disc on the shaft relative to the probe, thus varying the angle positions for the switching on and off of the data acquisition. However, as the angle $\Delta\theta$ between the slots remains fixed and the probe radius R is constant it follows for any test and disc orientation, that the acquired signal record $E(t)$ corresponds to a constant length S of the probe path. Finally, the apparatus was shielded by large boards placed around it, to prevent convection currents in the room affecting the hot-wire signal from the swinging arm probe.

The investigated velocity range can be varied by modifying the slot angle $\Delta\theta$ and probe radius R . In our case the required velocity range was approximately 1–6 m/s and this range was achieved by selecting $\Delta\theta = 180^\circ$ and R (Fig. 2) was evaluated as 768 mm for the tests quoted in the results Sect. 4. The corresponding probe path S was evaluated from $S = R \Delta\theta$.

3.1 Choice of release and switch angles

In order to evaluate the required calibration relationship two different velocity traces in the equivalent form of $E_2(t)$ and $E_2(t)$ are required. The motion of the swinging arm will to first order resemble a pendulum with frictionless bearings. If such a pendulum is released from a vertical position (release angle $\theta_R = 0^\circ$, Fig. 3a) it will complete a full circle and the velocity-angle curve will approximate a cosine shape curve as shown in Fig. 3b.

As explained we can start the data acquisition at any angle $\theta = \theta_s$ of the probe position, by changing the position of the disc on the shaft. The corresponding angle at which

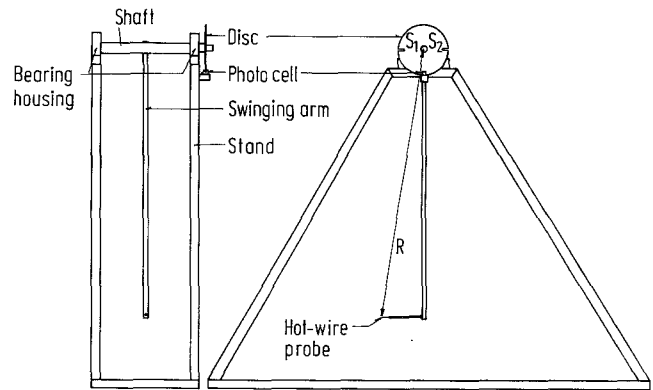


Fig. 2. Front and side view of swinging arm test rig

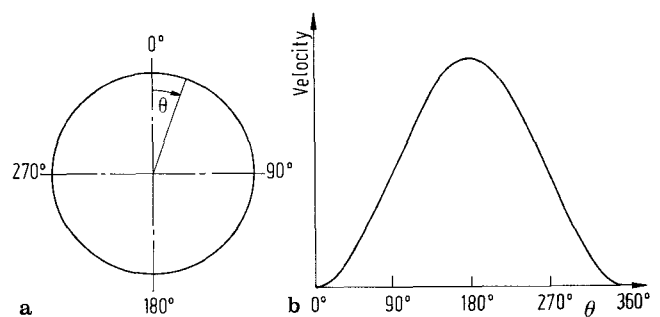


Fig. 3. a Angular notation, b related pendulum velocity-angle relationship

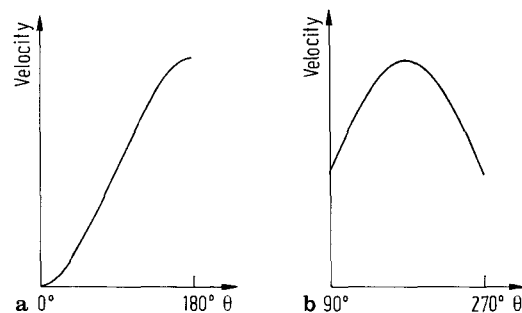


Fig. 4 a and b. Velocity-angle relationships, $\Delta\theta = 180^\circ$, $\theta_R = 0^\circ$; a $\theta_s = 0^\circ$, b $\theta_s = 90^\circ$

the data acquisition will stop will be $\theta_s + 180^\circ$. Consequently if we select a probe angle θ_R of 0° and if data acquisition starts at angles of respectively 0° and 90° then one will obtain velocity histories similar to those shown in Fig. 4. As required by the theory outlined in Sect. 2 these velocity time histories are sufficiently different, but they have the following disadvantages.

Firstly the velocity trace in Fig. 4a includes velocity points at or near zero velocity, which corresponds to natural convection conditions. This problem was overcome by starting the data acquisition at an angle θ_s , which was greater than θ_R .

Secondly the two velocity traces do not cover the same velocity range. It is noticed that the maximum velocity is obtained at $\theta = 180^\circ$ using a release angle $\theta_R = 0^\circ$. The corresponding velocity time history using a data acquisition start angle $\theta_s = 10^\circ$ will be referred to as test 1 (Fig. 5a). To obtain a second velocity time history which is different from test 1, but covering the same velocity range, it was necessary to combine signals from two test configurations. To achieve the maximum velocity, the probe in test 2 was also released at $\theta_R = 0^\circ$, but to obtain a different velocity time history the start angle θ_s was selected as 90° (Fig. 5b). This test covered the higher velocity range. To include the lower velocity range a test 3 was carried out using a release angle $\theta_R = 80^\circ$ and a start angle $\theta_s = 10^\circ$ (Fig. 5c). In this way tests 2 and 3 will cover the same velocity range as test 1. As outlined in the theory, the required two types of tests have to correspond to the same distance travelled by the probe. Denoting these two types of tests as I and II, we can identify test I as test 1 + test 1 and test II as test 2 + test 3.

4 Results

To demonstrate the repeatability and validity of the proposed method, results are presented for three different swings for each of the test configurations 1, 2, and 3 specified in Sect. 3.1. These individual tests will be identified as test 1A, 1B, 1C; test 2A, 2B, 2C, and test 3A, 3B, 3C. From these individual tests the required double swing tests I (1 + 1) and II (2 + 3) were evaluated in the form:

Test A: IA: test 1A + test 1B IIA: test 2A + test 3A
 Test B: IB: test 1B + test 1C IIB: test 2B + test 3B
 Test C: IC: test 1A + test 1C IIC: test 2C + test 3C.

The measured time history plots for $E(t)$ for the test IA and IIA are shown in Fig. 6, and the corresponding variation in $Y(n)$ Eq. (9) is illustrated in Fig. 1. The corresponding optimum value of $n \cong 0.44$ was obtained using the iterative procedure outlined in Sect. 2. Having evaluated n the corresponding value of B (in Eq. (2)) was obtained from Eq. 5. The corresponding velocity time history for test A is shown in Fig. 7, from which the velocity range covered is seen to be approximately 1.2–6.5 m/s.

The results obtained from the three tests A, B, C are shown in Table 1. To illustrate the difference between the three curvefits corresponding to test A, B and C two evaluations will be presented. Firstly, setting $V = 1$ m/s and 6 m/s it was evaluated that the variation between the corresponding calculated voltages for the three methods is

$$1 \text{ m/s: } E = 1.5013 \text{ V} \pm 0.0013 \text{ V}$$

$$6 \text{ m/s: } E = 1.8374 \text{ V} \pm 0.0007 \text{ V}$$

i.e., the observed uncertainty in E is about ± 1 mV. The corresponding velocity uncertainty can be estimated by assuming that one of the tests (e.g. test A) represents the correct response. For each selected velocity V this enables the evalu-

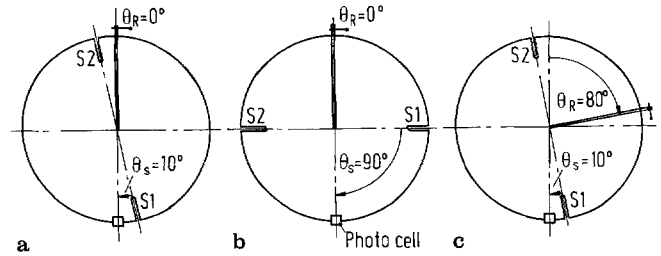


Fig. 5 a–c. Test configurations: **a** test 1: $\theta_R = 0^\circ$, $\theta_s = 10^\circ$, **b** test 2: $\theta_R = 0^\circ$, $\theta_s = 90^\circ$, **c** test 3: $\theta_R = 80^\circ$, $\theta_s = 10^\circ$

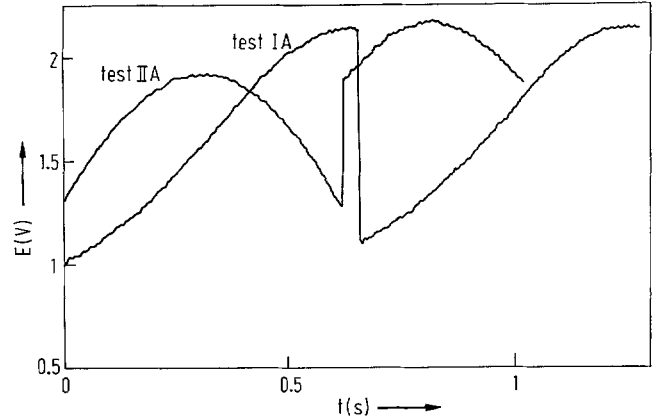


Fig. 6. Time history plots for $E(t)$ for tests IA and IIA

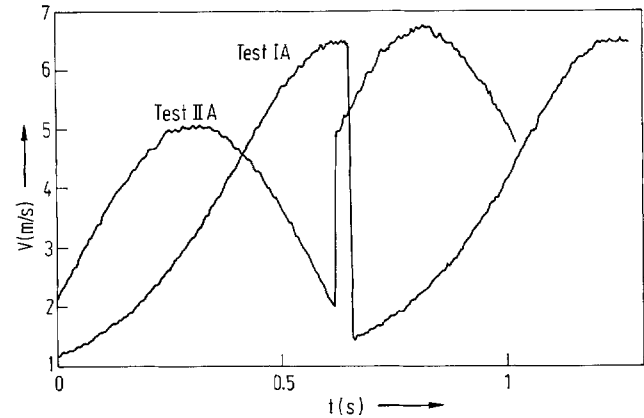


Fig. 7. Velocity time history plots for tests IA and IIA

Table 1. Calibration constants, Eq. (2), for three separate tests

Test	A	B	C
A	1.3087	0.9408	0.4399
B	1.3124	0.9432	0.4363
C	1.3112	0.9445	0.4365

ation of the corresponding voltage value E , which can then be used as input to test B and C to evaluate the corresponding V_B and V_C . Such a comparison for $V_A = 1, 3$ and 6 m/s is shown in Table 2, from which it is observed that the maximum variation is 0.03 m/s (or 0.5% at 6 m/s). This accuracy

Table 2. Comparative velocity evaluation using test A as a reference

V_A (m/s)	V_B (m/s)	V_C (m/s)
1	0.985	0.985
3	2.993	2.987
6	6.029	6.014

compares favourably with both the "sled" motion and the dynamic calibration methods.

For comparison a conventional calibration was also carried out in the velocity range 1 to 6 m/s using the DISA nozzle calibration test facility. The details of the test procedure is described in Khan et al. (1987) and will not be repeated here. However, it must be emphasised that the velocity at the probe position was evaluated from the pressure drop ΔP over the nozzle, and at these low velocities the value of ΔP is very small ($\Delta P < \sim 1$ mm H₂O). As the values of ΔP were measured with a Betz manometer (accuracy ± 0.02 mm H₂O), the uncertainty in the velocity evaluation will therefore be rather high. Consequently, the classical method cannot be used as a reliable reference in this low velocity range, but the results are included for general comparison. The nozzle calibration results gave the following values of the calibration constants:

$$A = 1.323, \quad B = 0.937, \quad n = 0.425.$$

These values are sufficiently close to the swinging arm results to confirm the validity of the swinging arm results.

Having demonstrated a high repeatability of the new method, it should prove a useful and reliable method for hot-wire and hot-film calibration at low velocities.

5 Conclusion

The results presented in this paper have clearly demonstrated the reliability of the proposed integral swinging arm calibration method. The test rig itself is very simple and the calibration experiments were easy to carry out. Also with access to modern computational facilities no difficulties should be experienced in implementing the required data acquisition and analysis procedures. This method should therefore prove to be a simple and reliable method for low velocity calibration of anemometer probes.

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